TIME RESPONSE ANALYSIS

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Introduction

- Time response of the system is defined as the output of a system when subjected to an input which is a function of time.
- Time response analysis means subjected the control system to inputs that are functions of time and studying their output which are also function of time.

TIME RESPONSE

- A control system generates an output or response for given input.
- The input represents the desired response while the output is actual response of system.
- Ex. Elevator

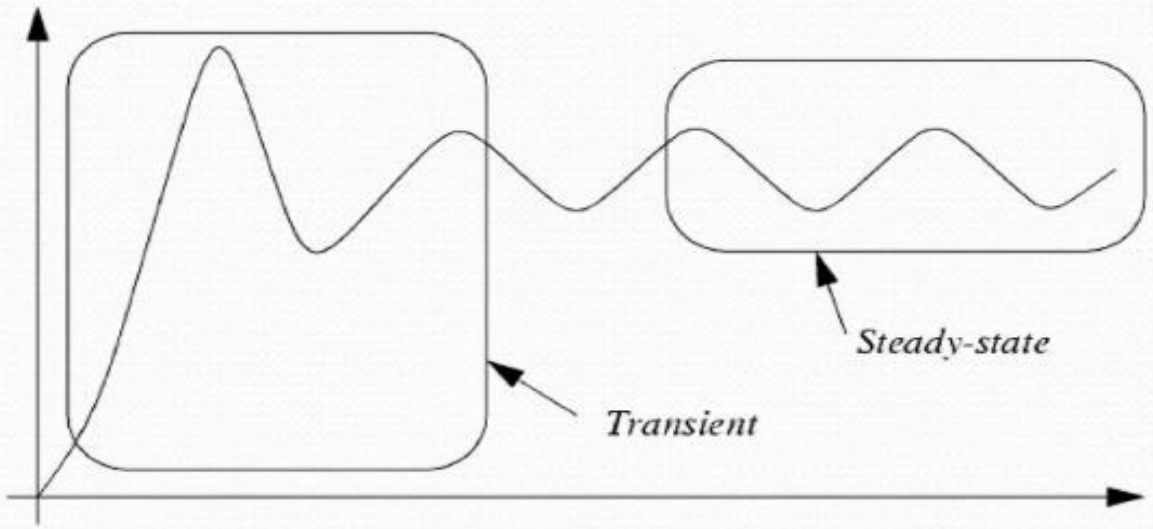
TIME RESPONSE

- As defined earlier, time response is the response of control system as a function of time.

- The time response analysis is divided into two parts

- **i)** the output is changing with respect to time. **(transient response)**

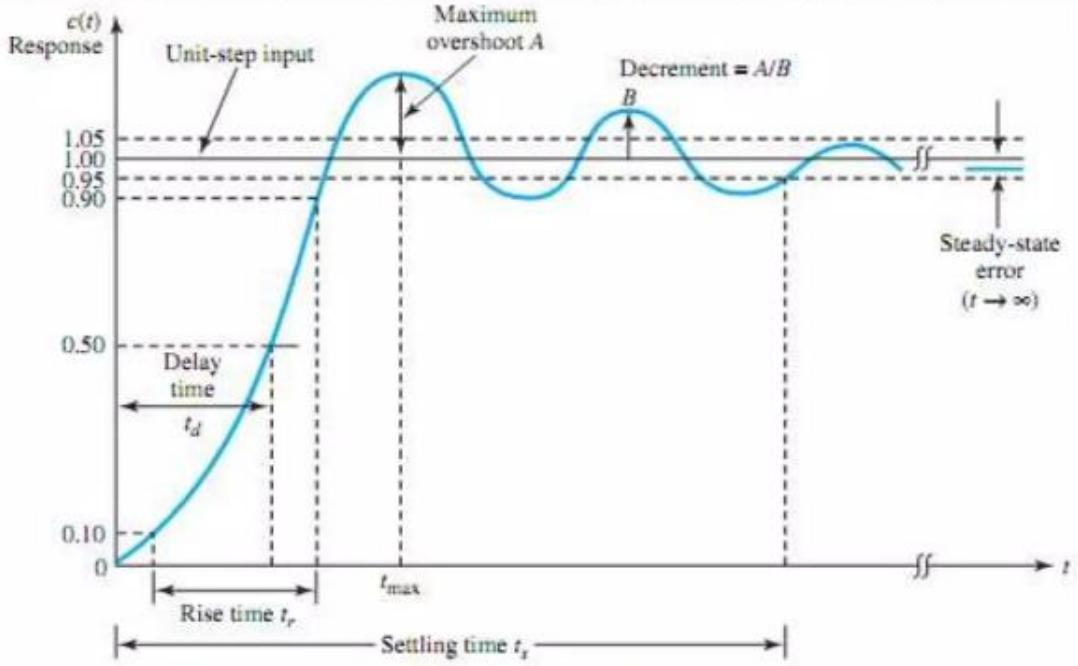
- **ii)** the output is almost constant. **(steady state response)**



Note: the transient response is predicted with the homogeneous solution. The steady state response in mainly predicted with the particular solution, although in some cases the homogeneous solution might have steady state effects, such as a non-decaying oscillation.

TIME RESPONSE

- So that the total time response, Ct(t) followed by the steady state response Css(t).
  - C(t) = transient + steady state response
  - C(t) = Ct(t) + Css(t)



Typical unit-step response of a control system.

INPUT SUPPLIED TO SYSTEM

- For time response analysis of control systems, we need to subject the system to various test inputs.
- Test input signals are used for testing how well a system responds to known input.
- Some of standards test signals that are used are:
  - Inpulse
  - Step
  - Ramp
  - Parabola
  - Sinusoidal

INPUT SUPPLIED TO SYSTEM

IMPULSE INPUT

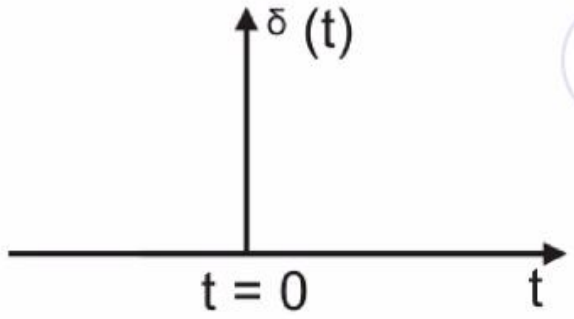
•It is sudden change input. An impulse is infinite at t=0 and everywhere else.

• $r(t) = \delta(t) = 1 \qquad t = 0$

$\qquad\qquad = 0 \qquad t \neq 0$

In laplase domain we have,

• $L[r(t)] = 1$



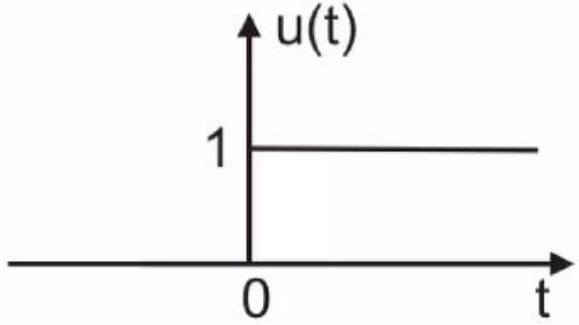
STEP INPUT

•It represents a constant command such as position. Like elevator is a step input.

•$r(t) = u(t) = A \qquad t \geq 0$

$\qquad\qquad = 0 \qquad$ otherwise

$L[r(t)] = A/s$



RAMP INPUT

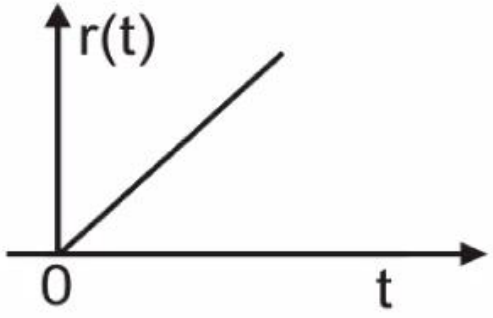
• this represents a linearly increasing input command.

•$r(t) \quad = At \qquad t \geq 0$,Aslope

$\qquad\quad = 0 \qquad t < 0$

$L[r(t)] = A/s^2$

A= 1 then unit ramp



INPUT SUPPLIED TO SYSTEM

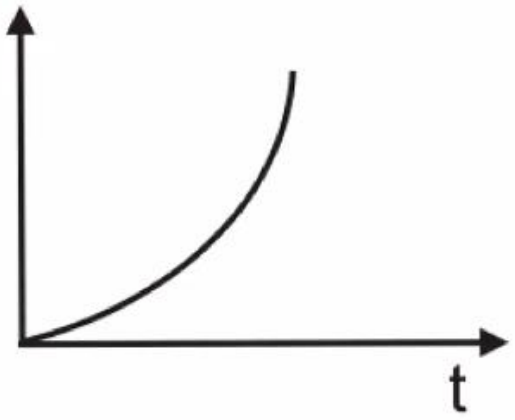
PARABOLIC INPUT

•Rate of change of velocity is acceleration. Acceleration is a parabolic function.

• $r(t) = At^2/2 \quad t \geq 0$

$= 0 \quad t < 0$

$L[r(t)] = A/s^3$



SINUSOIDAL INPUT

•It input of varying and study the system frequently response.

• $r(t) = A\sin(wt) \quad t \geq 0$

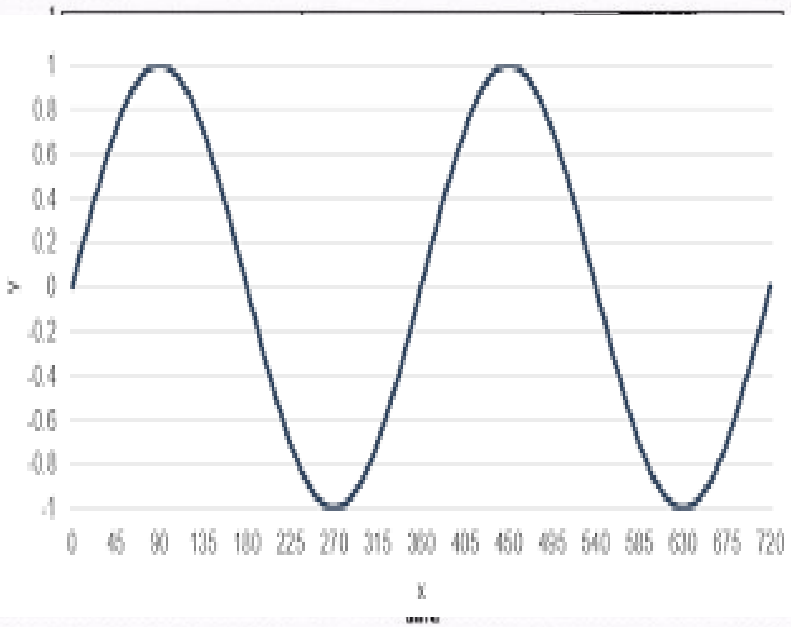


Figure 4: Simulink simulation of a first-order transfer function.

Order system

The order of the system is given by the maximum power of **s** in the denominator polynomial, Q(s)

Here, $Q(s) = a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \ldots\ldots\ldots\ldots + a_{n-1}s + a_n$.

Now, **n** is the order of the system

When n = 0, the system is zero order system.

When n = 1, the system is first order system.

When n = 2, the system is second order system and so on.

The numerator and denominator polynomial of equation (2.10) can be expressed in the factorized form as shown in equation (2.11).

$$T(s) = \frac{P(s)}{Q(s)} = \frac{(s+z_1)(s+z_2)\ldots\ldots(s+z_m)}{(s+p_1)(s+p_2)\ldots\ldots(s+p_n)} \qquad \ldots\ldots(2.11)$$

where, $z_1, z_2, \ldots\ldots z_m$ are zeros of the system.

$p_1, p_2, \ldots\ldots p_n$ are poles of the system.

Now, the value of **n** gives the number of poles in the transfer function. Hence the order is also given by the number of poles of the transfer function.

Introduction - First order system

- The first order system has only one pole.

$$\frac{C(s)}{R(s)} = \frac{K}{Ts+1}$$

- Where

  K is the D.C gain and

  T is the time constant of the system.

- Time constant is a measure of how quickly a 1st order system responds to a unit step input.
- D.C Gain of the system is ratio between the input signal and the steady state value of output.

First order system

- Examples:
- The first order system given below. $$G(s) = \frac{10}{3s+1}$$

- D.C gain is 10 and time constant is 3 seconds.

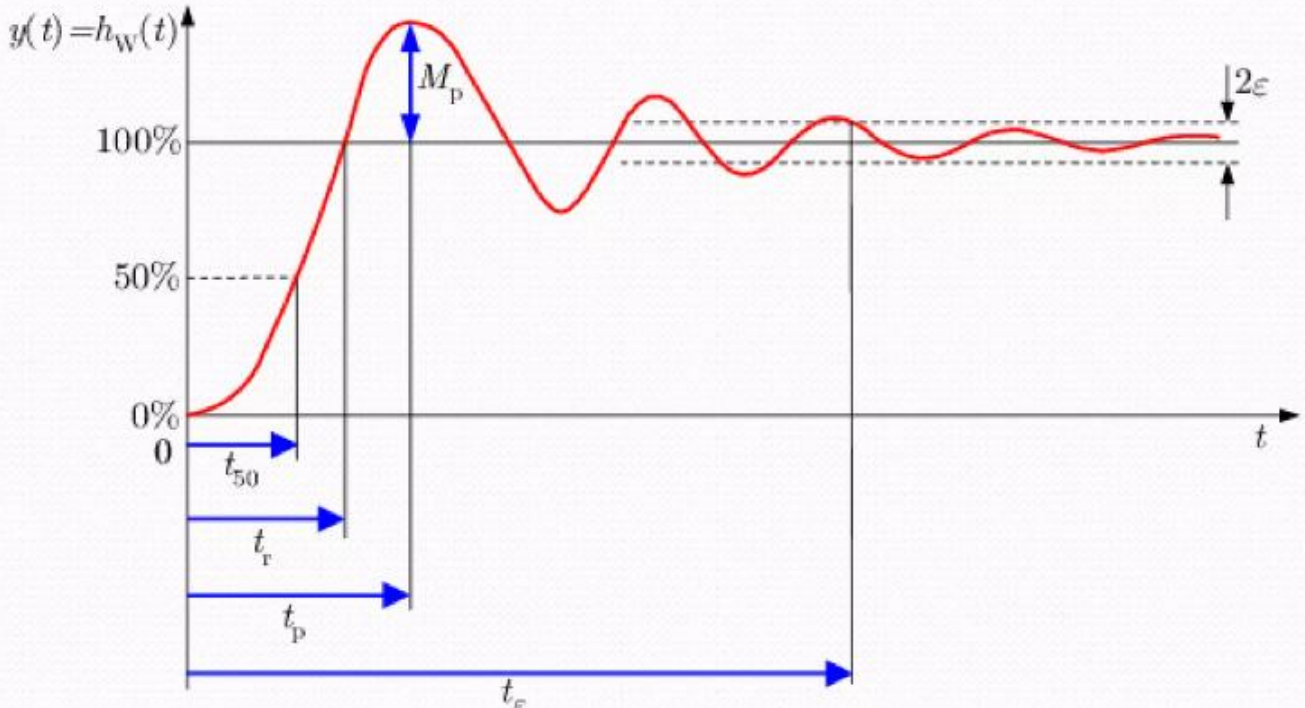
- For the following system

$$G(s) = \frac{3}{s+5} = \frac{3/5}{1/5s+1}$$

- D.C Gain of the system is 3/5 and time constant is 1/5 seconds.

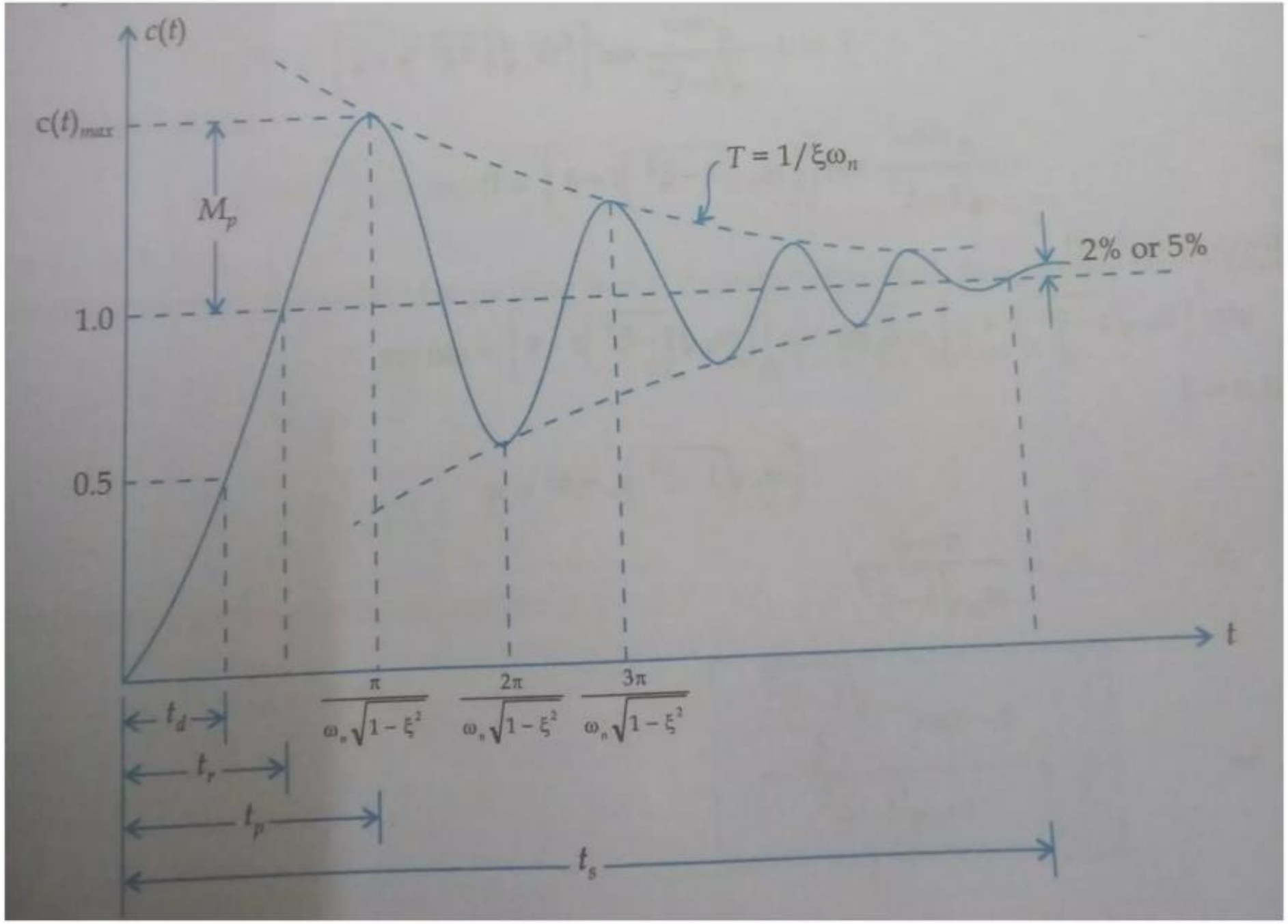
TIME RESPONSE SPECIFICATIONS

- The *maximum overshoot* is the magnitude of the overshoot after the first crossing of the steady-state value (100%).
- The *peak time* is the time required to reach the maximum overshoot.
- The *settling time* is the time for the controlled variable first to reach and thereafter remain within a prescribed percentage of the steady-state value. Common values of are 2%, 3% or 5%.
- The *rise time* is the time required to reach first the steady-state value (100%).



TIME RESPONSE SPECIFICATIONS

- Specifications for a control system design often involve certain requirements associated with the time response of the closed-loop system.
- The requirements are specified by the behavior of the controlled variable or by the control error on well defined test signals.
- The most important test signal is a unit step on the input of the control system and requirements are placed on the behavior of the controlled variable


c(t)
c(t)max
Mp
1.0
0.5
T = 1/ξωn
2% or 5%
t
π / ωn√(1 − ξ²)
2π / ωn√(1 − ξ²)
3π / ωn√(1 − ξ²)
td
tr
tp
ts

STEADY STATE RESPONSE

- The steady state response is that part of the output response where the output sognal remails constant.
- The parameter that is important in this is the steady state error(Ess)
- Error in general is the difference between the input and the output. Steady state error is error at $t \rightarrow \infty$

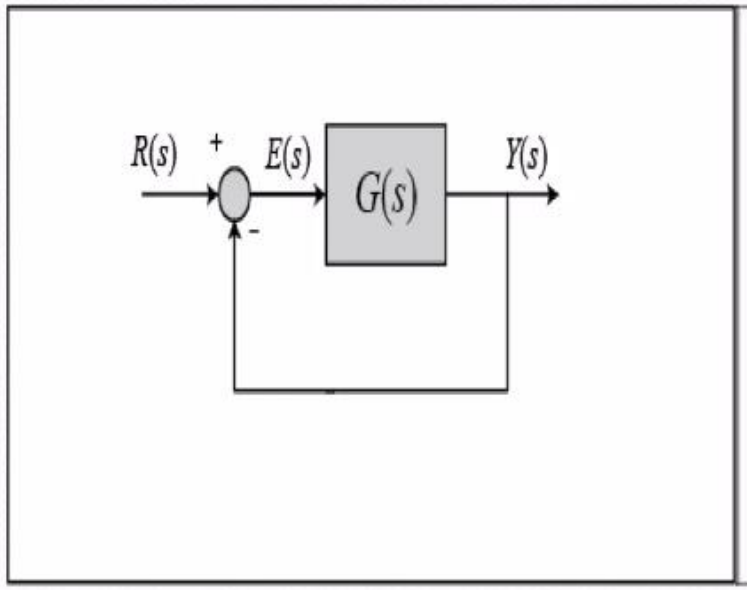
$$E_{ss} = \lim_{t \to \infty} e(t) = \lim_{s \to 0} sE(s)$$

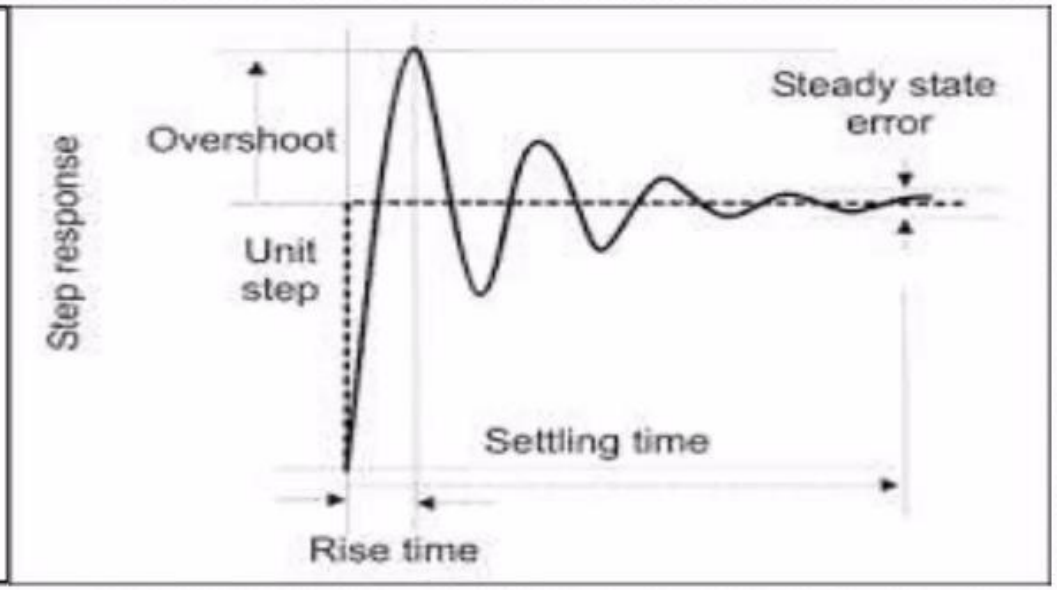
STEADY STATE ERROR

- Static error coefficient
- The response that remain after the transient response has died out is called steady state response
- The steady state response is important to find the accuracy of the output.
- The difference between the steady state response and desired response gives us the steady state error.
  - Kp = positional error constant
  - Kv = velocity error constant
  - Ka = acceleration error constant
- These error constant called as static error co efficient. they have ability to minimize steady state error.

STEADY STATE ERROR

Input	$e(\infty)$ Error	Type 0		Type 1		Type 2	
		K	$e(\infty)$	K	$e(\infty)$	K	$e(\infty)$
Step	$\frac{1}{1+K_p}$	$K_p = C$	$\frac{1}{1+K_p}$	$K_p = \infty$	0	$K_p = \infty$	0
Ramp	$\frac{1}{K_v}$	$K_v = 0$	∞	$K_v = C$	$\frac{1}{K_v}$	$K_v = \infty$	0
Parabola	$\frac{1}{K_a}$	$K_a = 0$	∞	$K_a = 0$	∞	$K_a = C$	$\frac{1}{K_a}$





LIMITATIONAS OF TIME DOMAIN ANALYSIS

- Control system analysis is carried out in either time domain or frequency domain. The domain of analysis depends largely on the design requirements.
- he analysis in the frequency domain is very simple and quick. Stability determination using a frequency response plot can be done in very quick time with no effort.
- In time domain analysis, the analysis becomes cumbersome for systems of high order. In frequency domain analysis, the order has a little effect on the time or effort of analysis.